Project Design Phase
Proposed Solution Template

Date	12 October 2026
Team ID	SWUID2026403219
Project Name	IMovies - Online Movie, Event & Entertainment Ticket Booking System
Maximum Marks	2 Marks

Proposed Solution for IMovies

S. No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Movie-goers and event fans struggle to find shows in one place, pick seats reliably and get quick confirmations, while manual counters and scattered apps cause queues, double bookings, unclear cancellations and delayed refunds. Cinemas and organizers lack simple tools to manage shows, pricing and live occupancy, which leads to empty seats and lost revenue.
2	**Idea / Solution Description**	IMovies is a full-stack web application (React frontend + Spring Boot REST backend + PostgreSQL/MySQL) where customers search by city, choose a cinema and showtime, select seats on an interactive map, add food, apply coupons, pay through a mock gateway and receive a QR e-ticket. Seats are locked for 10 minutes during checkout so two users can never book the same seat. Cinema managers, event organizers and admins manage shows, events, offers, refunds and analytics through role-based dashboards.

3	**Novelty / Uniqueness**	- Real-time seat map with atomic seat locking and automatic release on timeout - Single platform for movies, concerts, sports, plays and workshops - Policy-based cancellation (up to 30 minutes before show) with refund preview and tracking - Signed QR tickets that staff can verify and mark as used - Basic recommendation engine using bookings, genres, languages and ratings - Live occupancy and revenue analytics for cinemas and organizers

S. No.	Parameter	Description
4	**Social Impact / Customer Satisfaction**	- Saves time by removing queues and enabling booking from any device - Fair seat allocation and transparent pricing, fees and refunds - Gives local cinemas and small event organizers affordable digital reach - Reviews and ratings help audiences choose with confidence - Timely email and in-app reminders reduce missed shows
5	**Business Model (Revenue Model)**	- Convenience fee on every ticket booked - Commission from cinemas and organizers on event ticket sales - Paid featured listings and promotional banners - Sponsored offers and coupons from brands and food partners - Premium organizer plans with advanced analytics

6	**Scalability of the Solution**	- Stateless REST backend that scales horizontally behind a load balancer, with Redis for shared seat locks and caching - Designed to grow to 100,000 users, 500 cinemas and 50,000 bookings per day with indexing and pagination - Modular architecture for adding real payment gateways, SMS and push notifications - Future mobile apps, dynamic pricing, loyalty programs and multi-language support